CausalPilot: Evidence-Gated Autonomy for Agentic Business Experimentation

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Independent research prototype
https://github.com/appleweiping/causalpilot

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Abstract

Language-model agents can recommend business actions before the evidence supports acting. CausalPilot studies a narrower question than general agent accuracy: *when should an agent be allowed to experiment, ask a human, deploy to a segment, pause, roll back, or abstain?* We introduce TRACE, a trajectory-aware gate that keeps separate model, causal, data, and environment risk indices and combines them with deterministic statistical tools, budgets, and an explicit state machine. A seeded fashion-recommendation simulator makes harmful actions and regret observable; adapters support offline Open Bandit and corrected Criteo Uplift data without relabeling logged evidence as a live experiment. In a controlled 50-seed engineering benchmark (600 deterministic policy runs), TRACE made 2 false-deployment episodes (4%; Wilson 95% CI 1.1%–13.5%) at 26% mean coverage, while a no-TRACE point-estimate proxy made 76% at 80% coverage. Because coverage differs and several "LLM" comparators are disclosed decision-rule proxies, this is a system sanity check, not evidence of industrial superiority. The deliverable includes a FastAPI engine, a semantic 3D control room, reproducible artifacts, tests, and an in-memory append-only transition trace.

1 Problem and motivation

Business agents operate under a different failure model from question answering. A fluent but unsupported recommendation can become an irreversible allocation of traffic, budget, or customer treatment. Verbal confidence is especially weak when uncertainty comes from tool failures, selection bias, limited overlap, multiple segments, stale evidence, or a missing business constraint.

This project asks:

> Under fixed experiment-exposure and human-attention budgets, can a trajectory-aware risk gate allocate action rights more safely than direct recommendations, tool use without a gate, or a fixed causal pipeline?

The question connects recent work on agent uncertainty, abstention, causal agents, and experimental design. Agent-UQ research argues that uncertainty must cover heterogeneous tool observations and evolve over a trajectory [1]. AgentAbstain identifies post-hoc refusal, in which an agent acts irreversibly before recognizing that it should have abstained [2]. CausaLab finds that even strong models can stop experimentation prematurely and can predict correctly while recovering the wrong mechanism [3].

The project is also adjacent to two works from Tianshu Sun's research group. KP-Agent combines tools, memory, and a contextual bandit for keyword pruning; its reported gains are historical-data simulation results, not online causal estimates [4]. SEED represents human–agent workflows as typed actor-flow graphs and includes escalation and governance, while explicitly limiting its evidence to a feasibility contrast [5]. CausalPilot complements these works by implementing and evaluating the evidence conditions under which authority changes hands.

Hypotheses. H1 tests harmful deployment at matched coverage; H2 tests whether probe and human-query actions buy useful information; H3 tests whether trajectory risk predicts unsafe actions better than verbal confidence; H4 removes dynamic risk, probing, human queries, and value-of-information rules. The current artifact is sufficient for a mechanism check, but not yet for a conclusive matched-coverage or learned-calibration result.

2 TRACE design

TRACE stands for **T**rajectory **R**isk and **A**ction **C**ontrol for **E**xperimentation. The coordinator may parse goals, form hypotheses, propose tools, and write a short rationale. It cannot calculate statistical claims or commit a deployment. Deterministic code owns those operations.

2.1 Four risk indices

At step t, TRACE stores

$$R_t = (R_{\text{model}}, R_{\text{causal}}, R_{\text{data}}, R_{\text{environment}}),$$
$$R_{\text{overall}} = \max R_t.$$

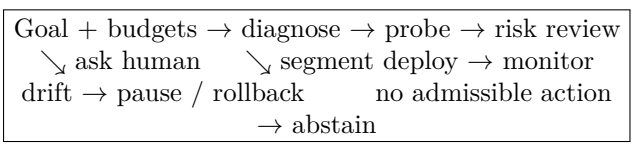
The maximum is a conservative engineering policy, not a learned optimum. Values are called *risk indices*, not probabilities, until a held-out reliability experiment supports calibration.

- **Model risk**: invalid goals or tool arguments, contradictory plans, missing evidence references, and illegal transitions.
- **Causal risk**: unknown assignment, poor identification, intervals crossing a harm boundary, estimator disagreement, or failed refutation.
- **Data risk**: sample-ratio mismatch (SRM), imbalance, weak overlap, low effective sample size, low precision, and multiplicity.
- **Environment risk**: covariate drift, out-of-support segments, stale evidence, changing constraints, and tool failure.

Each update records the before/after value, reason code, and evidence. A model cannot override schema failure, unsupported segments, or critical drift.

2.2 Actions and state machine

The API exposes eight actions: diagnose, run a randomized probe, ask one structured human question, deploy one segment, deploy all, pause a segment, roll back, and abstain. Preconditions form an inspectable authorization table. For example, deploy-all requires randomized evidence, no active drift, all segment intervals and diversity guardrails to pass, and every risk index to be at most 0.2. Pause requires a currently deployed target. Invalid actions return HTTP 409 and create no commit.



Reset creates a new mission identifier and links it to its parent; it does not overwrite the original in-memory history. The UI and server-sent events consume the episode snapshot; the benchmark reuses the same pure policy gate while keeping simulation truth in a scorer-only object.

2.3 Information and commit gates

A segment is admissible only if its lower 95% effect bound is positive, its diversity guardrail is above -2%, all four risk indices pass frozen thresholds, and overlap/integrity gates pass. Missing business value remains a vector rather than being converted to invented profit.

When nothing is deployable, the policy ranks bounded information actions with an approximate expected value of sample information,

$$\text{net EVSI}(a) = E[\max V \mid \text{observe}(a)] - \max V - \text{cost}(a).$$

If no information action has positive value or the budget is exhausted, the correct terminal result is abstention.

3 Implementation

The backend is an API-key-free Python/FastAPI service. It separates Pydantic contracts, the episode engine, the scenario generator, 13 deterministic statistical/validation tools, and a thread-safe in-memory store. Tools cover descriptive checks, randomized uplift and intervals, segment effects, SRM, balance, propensity overlap, two-proportion power, bootstrap sign stability, drift/PSI, guardrails, off-policy evaluation, and EVSI. Tool preconditions fail explicitly.

The Next.js/TypeScript interface is a "market control room," not a chat window. React Three Fiber renders fixed-seed user clusters. Cluster size maps to sample size, effect/risk maps to color, uncertainty maps to a halo, and particle batches map to simulated traffic. The interface also provides a two-dimensional table, text labels, keyboard controls, and reduced-motion behavior. If the API is unavailable, an explicitly labeled seeded simulator preserves the demo; it does not impersonate a remote model.

The flagship mission requests increased fashion recommendation conversion while limiting relative diversity loss to 2% under a 30,000 simulated-exposure budget. Historical selection bias blocks the direct proposal. A randomized probe reduces causal risk;

injected new-cohort drift pauses the affected segment; the system asks for one cost constraint; eligible segments deploy while others remain held or abstained.

3.1 Audit and reproducibility contract

Every episode event is append-only within one process and includes a sequence number, state transition, public action, and structured details. The mission-start event fixes evidence mode, fixture and configuration fingerprints, estimator, and gate thresholds. The log supports checking action order and state authorization; it is not a cryptographic ledger, durable event store, or complete replay proof. It does not persist private chain-of-thought. Reset starts a linked child episode so a failed trajectory remains available until process exit.

The repository treats generated results as build products with source lineage. A benchmark artifact records evidence mode, fixture fingerprint, methods, seeds, gate and detector settings, cost model, bootstrap seed, per-run decisions, and summary intervals. A generation script reads the canonical CSV artifact to create paper macros and the README chart; it fails when a required method or column is absent. CI reruns the default 50-seed artifact and rejects a byte-level diff. A smaller run is available only as an explicitly labeled smoke command.

4 Evidence modes and evaluation

Three badges prevent a visually impressive replay from overstating its evidence:

1. `SIMULATED GROUND TRUTH`: action-policy testing with known segment effects and injected drift;
2. `PUBLIC RCT --- OFFLINE`: corrected Criteo Uplift data for group/policy estimates, not individual counterfactual truth;
3. `LOGGED BANDIT --- OPE`: Open Bandit click logs with propensities, not conversion or profit.

Criteo's original 25,309,483-row release has a documented advertiser-related leak. The adapter therefore targets the corrected 13,979,592-row version and does not redistribute its CC BY-NC-SA data [6]. Open Bandit's small official logs support smoke testing; complete data remains external [7].

4.1 Controlled mechanism check

The canonical artifact pairs 12 policies on seeds 0–49. The named LLM baselines are deliberately labeled *deterministic proxies*; they are not executions of a named language model. Continuous summary intervals use a seed bootstrap with 2,000 resamples and seed 20260719; binary/count proportions use Wilson score intervals. Resource columns report probe exposures, human questions, tool actions, and zero model/API calls.

Policy	False deploy	Coverage	Regret
TRACE	4%	26%	0.0065
No TRACE	76%	80%	0.0031
Fixed pipeline	58%	66%	0.0042
Direct proxy	72%	77%	0.0037
Always deploy	100%	100%	0.0099
Always escalate	0%	0%	0.0170

Table 1: Means over 50 simulated episodes. False deploy is an episode-level ground-truth label. Regret is simulator policy regret.

TRACE's result is exactly "2 of 50 controlled episodes," not a guarantee. Its lower coverage is part of the result. The table does not establish H1 at matched coverage. It does show that the intended gate responds to drift and human-owned constraints: the no-dynamic-risk-update ablation reaches 38% false-deployment episodes; removing the human query yields 4% false-deployment episodes and a 26.5% constraint-violation rate. Removing the probe yields 0% coverage and raises mean regret to 0.0170. Raw per-seed runs and all intervals are checked in and regenerated byte-for-byte by CI.

5 Failure cases and limitations

The simulator rewards the designed gate. Drift and margin violations are first-class variables and TRACE receives the corresponding observations. This is useful for verifying mechanics but can overstate external validity. Future work needs preregistered scenario families written independently of the policy and real model executions under fixed budgets.

Risk is not yet calibrated probability. Bootstrap stability and diagnostics produce interpretable indices, but H3 requires a held-out unsafe-action dataset, reliability curves, Brier/ECE, and threshold selection before test episodes.

Coverage is not matched. Point policies deploy at different rates. A conclusive comparison needs threshold sweeps or constrained optimization that compares safety at equal coverage and value at equal risk.

Offline evidence cannot enact an adaptive experiment. Criteo and Open Bandit adapters currently provide schema checks and small-sample smoke paths, not a completed public-data validation. Even a full offline run could not show that CausalPilot assigned live traffic. The 90-second display therefore keeps its simulation badge visible.

Human value is scripted. The current question has structured decision impact, but no user study measures attention cost, answer quality, or organizational accountability.

The coordinator is deterministic by default. This makes the artifact reproducible and removes API-key risk during review. It does not evaluate language-model planning quality. An optional schema-bounded model coordinator is future work and must remain unable to bypass commit gates.

6 Conclusion

CausalPilot treats autonomy as an evidence-allocation problem. Its contribution is not the number of agents or tools, but an executable boundary between proposal and action: the agent can buy evidence, ask for a material constraint, deploy selectively, reverse course, or refuse. The current artifact verifies that boundary in a transparent simulator and deliberately limits its claims. The next research step is to calibrate trajectory risk on held-out failures and test matched-coverage policies on corrected public randomized data and, eventually, shadow-mode enterprise workflows.

References

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